Francesco BIANCONI
Curriculum vitae[1]

Office
Department of Engineering
Università degli Studi di Perugia
Via Goffredo Duranti 93
06125 Perugia, Italy
☎ +39 075 585 3706

Home
Via Luigi Catanelli 54
06135 Perugia, Italy

bianco@ieee.org
www.bianconif.net
bianconif

Personal details

Born: 17 July 1971, Perugia, Italy
Gender: Male
Citizenship: Italian

Education

University degrees

- Doctor of Philosophy, *Computer-aided Mechanical Design*
  Università degli Studi di Perugia, Perugia, Italy — Jan. 2001
- Master of Engineering, *Mechanical Engineering*
  Università degli Studi di Perugia, Perugia, Italy — Apr. 1997

Other courses

- *Applied Data Science Specialization*, IBM/Coursera — Jun. 2020

Languages

- English: Proficient
  - Cambridge CB CPE (grade B – 219/230) — Nov. 2021
  - Cambridge CB CAE (grade A – 81/100) — Apr. 2013
  - iBT TOEFL (104/120) — Dec. 2011
- Spanish: Advanced
  - DELE intermediate (95/100) — May 2006
- Italian: Native

[1] Last updated: October 22, 2026

IT & Programming skills

- Programming languages & VCS
  - C, C++, Java, Python and Git
- Digital typesetting & office automation
  - Microsoft Excel, Microsoft Word and LaTeX
- Scientific packages & data visualisation tools
  - Matlab, Mathematica and Tableau
- CAD/CAE
  - Autodesk AutoCAD, Autodesk Inventor and SolidWorks

Employment

- *Associate professor*, Department of Engineering, Università degli Studi di Perugia, Italy
  Jul. 2015–present
- *Assistant professor*, Department of Engineering, Università degli Studi di Perugia, Italy
  Nov. 2000–Jun. 2015
- *CAD Engineer* (internship), Umbra Cuscinetti SpA, Foligno (Italy) Oct.–Nov. 1998
- *Secondary school teacher* (Textile Technologies), Istituto Professionale Statale per l'Industria e L'Artigianato, Perugia (Italy) Dec. 1997–Jun. 1998
- *Secondary school teacher* (Civil and Industrial Plants), Istituto Tecnico Statale per Geometri, Perugia (Italy) Dec. 1997–Jun. 1998
- *Secondary school teacher* (Applied Mechanics), Istituto Tecnico Industriale Statale, Foligno (Italy)
  Jun.–Jul. 1997

Visiting positions

- *Academic visitor*, School of Mathematics, Computer Science and Engineering; City, University of London, United Kingdom Sep.–Dec. 2018
- *Academic visitor*, School Electronic Engineering and Computer Science; Queen Mary, University of London, United Kingdom Sep.–Dec. 2015
- *Visiting research fellow*, School of Computing Sciences; University of East Anglia, United Kingdom
  Oct.–Dec. 2010
- *Visiting researcher*, School of Industrial Engineering, Department of Engineering Design; Universidade de Vigo, Spain Jun.–Jul. 2009
  Sep. 2007
  Sep. 2006
  Sep. 2005

Teaching

Undergraduate courses

- *Graphical Communication for Engineering Design* (5 CFU[2]) + *CAD* (2 CFU), Università degli Studi di Perugia, Department of Engineering, BSc Engineering Management 2024–present
- *Machine Drawing* (10 CFU), Università degli Studi di Perugia, Department of Engineering, BSc Mechanical Engineering 2021–present
- *Technical Drawing* (5 CFU) + *CAD* (2 CFU), Università degli Studi di Perugia, Department of Engineering, BSc Industrial Engineering 2010–present
- *Computer skills* (2 CFU), Università degli Studi di Perugia, Department of Engineering, BSc Mechanical Engineering 2012–2022
- *Technical Drawing* (6 CFU) + *Computer skills* (2 CFU), Università degli Studi di Perugia, Department of Engineering, BSc Mechanical Engineering (curriculum Engineering Management) 2017–2021
- *Machine Drawing* (10 CFU) + *Computer skills* (2 CFU), Università degli Studi di Perugia, Department of Engineering, BSc Mechanical Engineering 2008–2012
- *Machine Drawing* (6 CFU) Università degli Studi di Perugia, Department of Engineering, BSc Mechanical Engineering 2000–2007
- *Technical Drawing* (4 CFU) + *Computer skills* (2 CFU) Università degli Studi di Perugia, Department of Engineering, BSc Energy Engineering and BSc Engineering Management 2005–2010
- *Technical Drawing* (5 CFU) Università degli Studi di Perugia, Department of Engineering, BSc Materials Engineering 2000–2004
- *CAD Laboratory* (2 CFU) Università degli Studi di Perugia, Department of Engineering, BSc Materials Engineering 2003–2005

Post-graduate courses

- *Product Design and Development* (6 CFU), Università degli Studi di Perugia, Department of Engineering, MEng Industrial Engineering 2009–2010
- *Design Methods of Industrial Engineering* (6 CFU), Università degli Studi di Perugia, Department of Engineering, MEng Mechanical Engineering 2008–2009

Post-master courses and continuous education

- *Computer-aided Design* (2,75 CFU), Università degli Studi di Perugia, Graduate Teacher Training Programme 2005
- *Computer-aided Design* (4 CFU), Università degli Studi di Perugia, Faculty of Engineering post-master course in Materials Engineering 2004

[2]Stands for *Credito Formativo Universitario*. This is the unit used in the Italian higher education system to measure and assess the student's work and effort. The minimum number of required credits is 180 for a Bachelor's and 120 for a Master's degree.

- *Technical Drawing I* (1,75 CFU), Università degli Studi di Perugia, Graduate Teacher Training Programme 2002
- *Technical Drawing II* (2 CFU), Università degli Studi di Perugia, Graduate Teacher Training Programme 2003

Short courses

- *Fundamentals of Engineering Drawing: Theory and Applications* (10h), FAIST Componenti S.p.A, Montone, Italy, Feb. 2011
- *Fundamentals of Pattern Recognition and Image Processing* (10h), Universidade de Vigo, Spain, Doctoral Programme in Environmental Engineering Jun. 2009
- *Introduction to Technical Drawing* (15h), Black & Decker Italia, Corciano, Italy Dec. 2008–Jan. 2009
- *Introduction to CAD/CAE* (10h), Master in Virtual Engineering, ITT s.c.a.r.l, Umbertide, Italy 2005
- *Introduction to Object-oriented Programming in C/C++*, Università degli Studi di Perugia, Italy, Doctoral Programme in Industrial Engineering Apr.–May 2004

Teaching in Erasmus interchange programmes

- *Expresión Gráfica* (Technical Drawing, 6h). BSc Energy Engineering and Mining, and BSc Energy Resources and Engineering, Universidade de Vigo, Spain Dec. 2013
- *Expresión Gráfica* (Technical Drawing, 6h). BSc Engineering of Industrial Technologies, BSc Management Engineering, BSc Electrical Engineering, BSc Mechanical Engineering, BSc Industrial Automation and Electronic Engineering, and BSc Industrial Chemical Engineering, Universidade de Vigo, Spain Jan. 2012
- *Expresión Gráfica* (Technical Drawing, 8h). BSc in Industrial Engineering, Universidade de Vigo, Spain Apr. 2019

Talks

- *Behind charts: A quick detour into data visualisation*. In *Topology by the River Ouse: AI, data shape and computation in Buckingham*. The University of Buckingham, Buckingham, United Kingdom (online) 27 March 2026
- *Il machine learning e il deep learning spiegati al medico nucleare* [*Machine learning and deep learning explained to the nuclear medicine specialist*]. Corso *Big Data, AI & oncologia nucleare: "Questo matrimonio s'ha da fare"?* [*Short course Big data, AI & nuclear oncology: shall we get them married?*]. The Italian Society of Nuclear Medicine, Molecular Imaging and therapy (AIMN), Rome, Italy 5 February 2026
- *Intelligenza artificiale e radiomica nella diagnostica per immagini: introduzione a strumenti e metodi* [*Artificial intelligence and radiomics for diagnostic imaging: an introduction to tools and methods*]. Corso *Tecniche di intelligenza artificiale nell'imaging: istruzioni per l'uso* [*Short course Artificial intelligence techniques for diagnostic imaging—how to*]. Perugia Hospital, Perugia, Italy 7 November 2025

- *IA e radiomica: uno sguardo d'insieme su strumenti e metodi* [*AI and radiomics: an overview of tools and methods*]. Corso *AI e radiomica in medicina nucleare: luci ed ombre* [*Short course on AI and radiomics in Nuclear Medicine: Lights and shades*]. The Italian Society of Nuclear Medicine, Molecular Imaging and therapy (AIMN), Rome, Italy 17 December 2024

- *Quale software scegliere* [*Which software to choose*]. Educational session *Radiomica come e perché* [*Radiomics how and why*], the 16th National Congress of the Italian Society of Nuclear Medicine and Molecular Imaging (AIMN), Milan, Italy 23 June 2024

- *Intelligenza artificiale applicata all'imaging medico nucleare: estrazione e analisi di features radiomiche* [*Artificial intelligence applied to Nuclear Medicine imaging: extraction and analysis of radiomics features*] (with B. Palumbo), webinar "Evoluzione tecnologica, digital health and Nuclear Medicine" [Technical evolution, digital health and Nuclear Medicine], Italian Society of Nuclear Medicine and Molecular Imaging (AIMN), The Health Technology Assessment (HTA) study group 26 Jan. 2024

- *A quick introduction to Radiomics*. Webinar Introduction to Radiomics for Nuclear Medicine, Italian Society of Nuclear Medicine and Molecular Imaging (AIMN), The AI-Radiomics study group 16 Jan. 2023;

- *Colour and texture descriptors for visual recognition: From the early years to convolutional networks*. Keynote lecture (online), 2nd International Conference on Computer Engineering and Science (CCES 2022), Islamic Azad University, Najafabad, Iran 20 Dec. 2022;

- *Intelligenza artificiale e radiomica: applicazioni in Medicina Nucleare* [*Artificial Intelligence and Radiomics: Applications in Nuclear Medicine*] (with B. Palumbo), Memorial Dr. Fausto Chionne "Dall'intelligenza artificiale alla medicina predittiva e di precisione: quali sviluppi in sanità" [From artificial intelligence to personalised and predictive medicine: Developments for the healthcare system; in memory of Dr. Fausto Chionne], Città della Pieve, Italy 13 Oct. 2022

- *Intelligenza artificiale e radiomica: uno sguardo d'insieme* [*Artificial intelligence and radiomics: an overview*]. Invited note at the round-table on Artificial Intelligence and Radiomics, the 15th National Congress of the Italian Society of Nuclear Medicine and Molecular Imaging (AIMN), Rimini, Italy 14 May 2022

- *Advances in modelling and analysis of the human body by computational imaging* (with G. Pascoletti). Keynote lecture (online), 2nd International Congress on Engineering Sciences and Multidisciplinary Approaches, Istanbul, Turkey 18 Sep. 2021

- *Radiomics in medical imaging: an overview*. Invited talk, IET Webinar Recent advances in Medical Image Analysis 25 Jun. 2021

- *Texture and colour descriptors for visual recognition: historical overview and applications to computer vision and robotics*. Keynote lecture (online), The 2020 International Conference on Control, Automation and Diagnosis (ICCAD'20), Paris, France 7 Oct. 2020

- *Role of artificial intelligence techniques (automatic classifiers) in molecular imaging modalities in neurodegenerative diseases* (with B. Palumbo). Invited talk, short course in Big Data, Radiomics & Artificial Intelligence; Italian Association for Medical Physics (AIFM), Reggio Emilia, Italy 15-16 Dec. 2017

- *Towards a procedural model for CAD data exchange*; 5th workshop on Design Tools and Methods in Industrial Engineering, Pisa, Italy, 21-23 Mar. 2005
- *Collaborative CAD modeling and construction of augmented CAD models*; 4th workshop on Design Tools and Methods in Industrial Engineering, Erice, Italy 29 Sep.–1 Oct. 2003
- *Approaches for integration of CAD/CAM/CAE systems*; 3rd Workshop on Design Tools and Methods in Industrial Engineering, Firenze, Italy 27-28 Jun. 2002
- *Interface-based methods for data exchange among CAx systems*; 2nd Workshop on Design Tools and Methods in Industrial Engineering, Perugia, Italy 5-6 Jul. 2001
- *Collaborative design and data exchange through STL files*; 1st Workshop on Design Tools and Methods in Industrial Engineering, Parma, Italy Sep. 2000

Seminars

- *Texture and colour descriptors for visual recognition: an overview of methods applications*. Doctoral programme in Industrial and Information Engineering, Università degli Studi di Perugia, Italy 23 Jun. 2021
- *Hand-designed descriptors vs. pre-trained convolutional networks: a comparison of two strategies for colour texture classification*. School of Mathematics, Computer Science and Engineering; City, University of London, United Kingdom 20 Nov. 2018
- *Texture description through histograms of equivalent patterns: A unifying Framework for LBP and related methods*. School of Computing Sciences, University of East Anglia, United Kingdom 9 Dec. 2015
- *Texture description through histograms of equivalent patterns: A unifying Framework for LBP and related methods*. School of Electronic Engineering and Computer Science; Queen Mary, University of London, United Kingdom 22 Sep. 2015
- *Introduction to computer vision*, School of Industrial Engineering, Universidade de Vigo, Spain 21 Jan. 2014
- *Fundamentals of pattern recognition and colour image analysis*, School of Industrial Engineering, Universidade de Vigo, Spain 27 Nov. 2012
- *Introduction to pattern recognition and computer vision with applications in the industry*, School of Industrial Engineering, Universidade de Vigo, Spain 24 Jan. 2012
- *Colour vision and pattern recognition*. School of Industrial Engineering, Universidad de Vigo, Spain 11 Jan. 2011
- *Automatic characterization of materials appearance through texture and colour analysis*. School of Computing Sciences, University of East Anglia, United Kingdom 15 Oct. 2010
- *Data exchange among CAD/CAM/CAE systems: problems and perspectives*. Università degli Studi dell'Aquila, Italy 16 Dec. 2004

Tutorials

- *Colour texture analysis and classification* (with C. Cusano and P. Napoletano), 5th Computational Colour Imaging Workshop (CCIW'17), Milan, Italy 29 Mar. 2017

Supervision of PhD dissertations

- M.U. Khan *Radiomics as a tool to differentiate benign vs. malignant lesions in different imaging modalities* Doctoral Programme in Industrial and Information Engineering, Università degli Studi di Perugia, Italy Apr. 2026
- R. Bello-Cerezo *Colour texture classification at the end of the 'early' years: hand-designed descriptors or pre-trained convolutional neural networks?* Doctoral Programme in Industrial and Information Engineering, Università degli Studi di Perugia, Italy Apr. 2019

Supervision of BSc and MSc theses

- Thirty BSc and MSc theses within the Department of Engineering, Università degli Studi di Perugia, Italy 2001–present
- Eight MSc theses within the School of Industrial Engineering, Universidade de Vigo, Spain 2007–2014

Administrative work

- Member of the research committee, Department of Engineering, Università degli Studi di Perugia, Nov. 2014–Nov. 2025
- Member of the internship committee, BSc and MSc in Mechanical Engineering, Università degli Studi di Perugia Nov. 2022–present
- Member of the quality assurance group, BSc and MSc in Mechanical Engineering, Università degli Studi di Perugia Jan. 2024–present
- Member of the steering committee of the Doctoral Programme in Industrial and Information Engineering, Università degli Studi di Perugia 2013–present
- Member of the steering committee of the Doctoral Programme in Industrial Engineering, Università degli Studi di Perugia 2003—2012

Research projects

As principal investigator

- *Analisi della geometria e della tessitura di modelli piani e volumetrici: metodi ed applicazioni* (*STARTLE – Shape and Texture Analysis of planaR and volumeTric models: mEthods and applications*). Fundamental research grants, Department of Engineering, Università degli Studi di Perugia, Italy. Amount granted: € 4000,00. 2022-2024

- *Caratteristiche di forma, colore e tessitura per l'analisi di immagini piane e volumetriche: metodi ed applicazioni* (*Shape, colour and texture features for the analysis of two- and three-dimensional images: methods and applications*). Fundamental research grants, Department of Engineering, Università degli Studi di Perugia, Italy. Amount granted: € 2.751,32. 2020-2021

As investigator/participant

- *MODUS-PRO MODellazione aUtomatica perSonalizzata di PRotesi e Ortesi [Automated personalised modeling of protheses and ortheses]*. Research call 2025 – innovative research projects, Fondazione Perugia (Italy). Overall budget: € 50.000,00 2026-2028

- *Innovation, digitalisation and sustainability for the diffused economy in Central Italy (VITALITY). Spoke 9: NANOMAT pole – Nanostructured materials and devices*. NextGenerationEU (National Recovery and Resilience Plan – NRRP) grant, Italian Ministry for University and Research (MUR) and the European Commission. Overall budget: € 18.482.071,99 2024-2025

- *3D printing and digital twinning ceramic restorations for dentistry (3DCer4Dent)*. National research grants (PRIN 2022), Italian Ministry for University and Research (MUR), Italy. Amount granted: € 306.790,00 (total), 86.589,00 (Università degli Studi di Perugia) 2023-2025

- *Artificial intelligence for Earth observation*. Fundamental research grants, Department of Engineering, Università degli Studi di Perugia, Italy. Amount granted: € 3000,00. 2021-2022

- *Algoritmi classici e di Machine Learning per lo sviluppo di modelli sperimentali "data driven" per applicazioni robotiche e per la classificazione di prodotti industriali* (*Traditional and Machine Learning algorithms for developing data-driven robotics applications and for automatic classification of industrial products*). Fundamental research grants, Department of Engineering, Università degli Studi di Perugia, Italy. Amount granted: € 5600,00. 2019-2020

- *Identificación basada en objetos de cultivos hortícolas bajo invernadero a partir de stereo imágenes del satélite WorldView-3 y series temporales de Landsat-8 – Ref. AGL2014-56017-R* (*Object-based identification of greenhouse horticultural crops through satellite stereo imagery from WorldView-3 and time series from Landsat-8*). Ministry of Economy and Competitiveness, Spain; Universidad de Almería, Spain. Amount granted: € 85.000,00. 2015-2018

- *BioMeTron: Un nuovo approccio integrato e multidisciplinare per lo studio, la gestione e la progettazioni di impianti biologici per l'energia e l'ambiente* (*A new multi-disciplinary, integrated approach for studying, designing and managing biological plants for energy and the environment*). Fundamental research grants, Department of Engineering, Università degli Studi di Perugia, Italy. Amount granted: € 14.000,00. 2015-2017

- *LIFE12 ENV/IT/000411 Enhanced material recovery and environmental sustainability for small scale waste management systems*. European Commission. 2013-2018

- *GEOEYE1-WV2: Generación de datos georeferenciados de muy alta resolución a partir de imágenes de los satélites GeoEye-1 Y WorldView-2 – ref. CTM2010-16573* (*Generation of high resolution geo-referenced data from GeoEye-1 and WorldView-2 satellite images*). Ministry of Science and Education, Spain; Universidad de Almería, Spain. 2011-2013

- *EFESO: Environmental Friendly Energy from Solid Oxide Fuel Cells*. Ministry of Economic Development, Italy. Total amount granted: € 10.922.360,00 2009-2011

- *Modeling, simulation and experimental evaluation of materials for Industrial and Civil Engineering*. Consortium for the development of the University District, Terni, Italy. Amount granted: € 73.000,00 2007-2008
- *Sistemi innovativi per la gestione e la condivisione delle informazioni di prodotto in ambienti CAx (Innovative systems for sharing and managing product data in CAx environments)*. Ministry of Education, University and Research; PRIN programme 2005, Italy. Amount granted: € 73.000,00 2007-2008

Other grants and fellowships for teaching, research and consultancy

- *FFABR: Finanziamento delle Attività Base di Ricerca* (*Funding for Basic Activities Related to Research*). National Agency for the Evaluation of Universities and Research Institutes, Italy. Amount: € 3000,00. 2017
- *Introduction to Engineering Drawing: Theory and applications*. Faist Componenti S.p.A., Montone (PG), Italy. Grant type: teaching and consultancy. Amount: € 3000,00. 2018
- *Introduction to Engineering Drawing: Theory and applications*. Faist Componenti S.p.A., Montone (PG), Italy. Grant type: teaching and consultancy. Amount: € 1500,00. 2017
- *Introduction to Engineering Drawing: Theory and applications*. Faist Componenti S.p.A., Montone (PG), Italy. Grant type: teaching and consultancy. Amount: € 1000,00. 2016
- *Introduction to Computer Vision*. Universidade de Vigo, Spain. Grant type: teaching. Amount: € 500,00. 2014
- *Fundamentals of pattern recognition and colour image analysis*. Universidade de Vigo, Spain. Grant type: teaching. Amount: € 500,00. 2013
- *Fundamentals of pattern recognition and colour image analysis*. Universidade de Vigo, Spain. Grant type: teaching. Amount: € 1200,00. 2012
- *Colour vision and pattern recognition*. Universidade de Vigo, Spain. Grant type: teaching. Amount: € 1200,00. 2011
- *Digital image processing and pattern recognition*. Universidade de Vigo, Spain. Grant type: teaching. Amount: € 1800,00. 2009
- *Introduction to Engineering drawing, materials and manufacturing processes*. Black & Decker Italia S.p.A., Corciano (PG), Italy. Grant type: teaching and consultancy. Amount: € 6000,00. 2008-2009
- *Characterization of the visual appearance of ornamental stone through combination of classifiers*. Universidade de Vigo, Spain. Grant type: visiting fellowship. Amount: € 1200,00. 2006
- *Characterization of the visual appearance of natural stones (marble and granite) through image processing and artificial intelligence*. National Research Council, Italy. Grant type: Short-term mobility fellowship. Amount: € 1.787,87. 2005
- *Design and optimisation of shelters*. O.M.C. Srl, Passignano (PG), Italy. Grant type: consultancy. Amount: € 5.760,00. 2005

Editorial work

- Guest Editor of the Special Issue 'Radiomics in cancer imaging: Theory and applications in solid tumours', *Cancers* (e-ISSN 2072-6694) closing Oct. 2025
- Academic Editor for *PLOS One* (e-ISSN: 1932-6203) Jun. 2020–Mar. 2024
- Academic Editor for *Applied Sciences* (e-ISSN: 2076-3417) Nov. 2019–Feb. 2022
- Guest Editor of the Special Issue 'Artificial intelligence in Image-Based Diagnostics of Oncological and Neurological Disorders 2.0', *Diagnostics* (e-ISSN 2075-4418) closed Dec. 2023
- Managing Guest Editor of the Virtual Special Issue 'Visual dEscriptors for TexturE Recognition: from Gabor to deep leArNing (VETERAN)', *Pattern Recognition Letters* (ISSN: 0167-8655) closed Sep. 2022
- Guest Editor of the Special Issue 'Automated Product Inspection for Smart Manufacturing', *Applied Sciences* (e-ISSN: 2076-3417) closed Aug. 2022
- Guest Editor of the Special Issue 'Artificial intelligence in Image-Based Diagnostics of Oncological and Neurological Disorders', *Diagnostics* (e-ISSN 2075-4418) closed Dec. 2021
- Guest Editor of the Special Issue 'Texture and Colour in Image Analysis', *Applied Sciences* (e-ISSN: 2076-3417) closed Apr. 2020
- Reviewer for:
  Applications in Engineering Science (3 manuscripts revised); *Applied Mathematics and Computation* (2); *Algorithms* (4), *Applied Sciences* (13), *Archaeometry* (1), *Automation in Construction* (3); *Autosoft Journal* (2), *Bioengineering* (2), *Biomedicines* (1), *Biomecules & Biomedicine* (1), *Biosystems Engineering* (2), *BMC Cancer* (1), *BMC Medical Imaging* (7), *Cancer Imaging* (3), *Cancer Investigation* (1), *Cancer Medicine* (2), *Cancer Management and Research* (1), *Cancers* (6), *Clinical and Translational Imaging* (3), *Clinical Lung Cancer* (1), *Computers and Electronics in Agriculture* (1), *Computers in Industry* (2), *Computer Methods and Programs in Biomedicine* (3), *Concurrency and Computation: Practice and Experience* (1), *Data in Brief* (1), *Diagnostics* (4), *Digital Signal Processing* (4), *Dyna-Colombia* (1), *Ecological Informatics* (1), *Electronics* (3), *Energy and Buildings* (1), *Entropy* (2), *Engineering Computations* (1), *Expert Review of Medical Devices* (2), *Expert Systems With Applications* (11), *EURASIP Journal on Image and Video Processing* (1), *European Journal of Nuclear Medicine and Molecular Imaging* (1), *European Radiology* (2), *Fibers and Polymers* (1), *Food Science and Applied Biotechnology* (1), *Frontiers in Medicine* (1), *Frontiers in Oncology* (2), *Future Generation Computer Systems* (1), *IEEE Access* (1), *IEEE Journal of Biomedical and Health Informatics* (1), *IEEE Signal Processing Letters* (2), *IEEE Transactions on Circuits and Systems for Video Technology* (1), *IEEE Transactions on Emerging Topics in Computational Intelligence* (1), *IEEE Transactions on Image Processing* (19), *IEEE Transactions on Medical Imaging* (1), *IEEE Transactions on Multimedia* (2), *IEEE Transactions on Signal Processing* (1), *IET Image Processing* (2), *IET Signal Processing* (2), *IET Computer Vision* (2), *Information* (1), *Información Tecnológica* (1), *Integrated Computer-aided Engineering* (1), *Intelligent Automation and Soft Computing* (1), *International Journal of Computer Applications in Technology* (1), *International Journal of Computer Vision* (1); *International Journal of Continuing Engineering Education and Life-Long Learning* (1), *International Journal of Cosmetic Science* (1), *International Journal on Interactive Design and Manufacturing* (4) *International Journal of Multimedia Information Retrieval* (1), *International Journal of Neural Systems*

(2), *Journal of Applied Remote Sensing* (1), *Journal of Cancer Research and Clinical Oncology* (1), *Journal of Computational Methods in Sciences and Engineering* (1), *Journal of Computer Assisted Radiology and Surgery* (1), *Journal of Digital Imaging* (1), *Journal of Electronic Imaging* (9), *Journal of King Saud University* (1), *Journal of Imaging* (8), *Journal of Imaging Science and Technology* (1), *Journal of Medical Imaging* (1), *Journal of Medical Imaging and Health Informatics* (9), *Journal of the Optical Society of America – A* (1), *Journal of Visualization* (3), *Journal on Computing and Cultural Heritage* (1), *Knowledge-Based Systems* (3), *Life* (1), *Machine Vision and Applications* (1), *Measurement* (2), *Medical Image Analysis* (2), *Microscopy Research & Technique* (1), *Modelling and Simulation in Engineering* (2); *Molecular Imaging and Biology* (2), *Multidimensional Systems and Signal Processing* (7), *Nanomaterials* (1), *Neurocomputing* (1), *Optical Engineering* (4), *Optics and Lasers in Engineering* (7), *Optik - International Journal for Light and Electron Optics* (1), *Opto-Electronics Review* (1), *Pattern Analysis and Applications* (1), *Pattern Recognition* (12), *Pattern Recognition Letters* (9), *PeerJ Computer Science* (4), *Photogrammetric Engineering and Remote Sensing* (2), *PlosONE* (6), *Quantitative Imaging in Medicine and Surgery* (5), *Radiation Oncology* (1), *Remote Sensing* (3), *Respiratory Research* (1), *Scientific Reports* (4), *Signal Processing* (1), *SN Applied Sciences* (2), *Surfaces & Coatings Technology* (1), *Sensors* (9), *Solar Energy Materials & Solar Cells* (1), *Superficies y Vacío* (1), *Symmetry* (4), *Technologies* (2), *Textile Research Journal* (1), *Theranostics* (1), *Translational lung cancer research* (3), *Turkish Journal of Agriculture and Forestry* (1).

- MDPI Cancers – Exceptional Reviewer (Q4, 2024)

Participation in the organisation of scientific events

Organising committees

- The 6th International Conference on Control, Automation and Diagnosis (ICCAD'22); Lisbon, Portugal. Role: co-publication chair Jul. 2022
- The 5th International Conference on Control, Automation and Diagnosis (ICCAD'20); Paris, France. Role: co-publication chair Oct. 2020
- *Texture Analysis, Classification and Retrieval (TAILOR)*. Workshop within the 25th International Conference on Pattern Recognition (ICPR); Milan, Italy. Role: workshop organiser Jan. 2021
- *Processing visual data in intelligent systems: methods and applications*. Special track within the 10th KES International Conference on Intelligent Interactive Multimedia: Systems and Services; Vilamoura, Portugal. Role: session organizer and chair Jun. 2017

Scientific and programme committees

- Heritechne Conference 2026
- EG-ICE International Workshop on Intelligent Computing in Engineering 2025
- International Conference on Image Processing and Vision Engineering (IMPROVE 2024, IMPROVE 2025)
- International Conference on Frontiers of Digital Signal Processing (CFDSP 2023)
- International Symposium on Mathematical and Computational Oncology, Virtual (ISMCO'21)

- International Conference on Computer Graphics and Digital Image Processing (CGDIP 2020, CGDIP 2021)
- International Conference on Frontiers of Digital Signal Processing (CFDSP 2021)
- The International Conference on Control, Automation and Diagnosis (ICCAD'20, ICCAD'21 – co-publication chair, ICCAD'23, ICCAD'24, ICCAD'25 – co-special session chair and ICCAD'26 – co-programme chair)
- First International Symposium on Information and Future Technologies (ISIFut 2020)
- 15th European Congress on Digital Pathology (ECDP 2019)
- 4th International Workshop on Interactive and Spatial Computing (IWISC 2019)
- KES International Conference on Intelligent Decision Technologies (IDT 2019)
- International Conference on Advances in Signal Processing and Artificial Intelligence (ASPAI 2019)
- International Conference on Big Data and Machine Learning (BDML 2018)
- International Conference Florence Heri-Tech (The future of Heritage Science and technologies, 2018, 2020 and 2022)
- International Conference on Advances in Image Processing (ICAIP 2018–2019)
- Computational Color Imaging Workshop (CCIW'17)
- KES International Conference on Intelligent Interactive Multimedia: Systems and Services (IIMSS-17, IIMSS-18)
- International Conference on Communication and Information Processing (ICCIP 2016–2020)
- 1st IEEE Italy Section Summer School – Advanced course for graduated students and industrial research (IEEESS 2015)
- Knowledge-Based and Intelligent Information & Engineering Systems (KES 2015–2017)
- Workshop Colour in Texture and Material Recognition (in conjunction with the 18th Int. Conf. on Image Analysis and Processing – ICIAP 2015)
- Portuguese Conference on Artificial Intelligence (EPIA 2015)
- International Conference on Systems (ICONS 2015–2024)
- International Conference on Modelling, Simulation, and Identification (MSI 2014, MSI 2016)
- World Conference on Information Systems and Technologies (WorldCIST'13–21)
- International Conference on Computer Graphics and Imaging (CGIM 2013)
- Irish Machine Vision and Image Processing Conference (IMVIP 2011, IMVIP 2014–2023)
- Imaging and Signal Processing in Healthcare and Technology (ISPHT 2011)
- Computational Photography (CPhoto 2011)

- Visualization Imaging and Image Processing (VIIP 2007–2009 and VIIP 2012)
- International Conference on Information Visualization (IV07–10, IV2011, IV2012, IV2013, IV2015 and IV2016)
- Applied Simulation and Modelling (ASM 2008, ASM 2009 and ASM 2012)
- International Joint Congress XVI ADM – XIX INGEGRAF (2007)

Other

- Webinar on *Quantitative imaging in oncology: The role of radiomics in solid tumours*, MDPI Cancers, 21 Sep. 2026 Role: chair
- ADM 2026 International Conference Role: session chair
- Webinar on *AI-radiomica su PET dinamica: verso una medicina di precisione nella distinzione infiammazione-neoplasia* [AI-radiomics on dynamic PET: towards precision medicine for differentiating inflammation vs. malignancy], the Italian Society of Nuclear Medicine and Molecular Imaging, 25 Feb. 2026 Role: moderator
- Webinar on *Intelligenza artificiale e PET nella diagnosi e gestione clinica delle vasculiti: nuove frontiere per la medicina* [Artificial Intelligence and PET for the diagnosis and clinical management of vasculitis: new frontiers for medicine], the Italian Society of Nuclear Medicine and Molecular Imaging, 12 Nov. 2025 Role: moderator
- ADM 2025 International Conference Role: reviewer
- ADM 2024 International Conference Role: reviewer and session chair
- The 27th International Conference on Pattern Recognition (ICPR 2024) Role: reviewer
- ADM 2023 International Conference Role: reviewer and session chair
- International Joint Conference on Mechanics, Design Engineering and Advanced Manufacturing (JCM 2022) Role: reviewer and session chair

Advisory and evaluation

- Member of the selection committee for one post-doctoral position (ref. CR-2026-06) in the Department of Engineering, Università degli Studi di Perugia, Italy
  Oct. 2026
- Member of the final evaluation committee for the degree of Doctor of Philosophy, Inter-university Doctoral Programme in Bioengineering and Medical-Surgical Sciences, Politecnico di Torino, Italy
  May 2026
- Member of the selection committee for one post-doctoral position (ref. IR-2026-02) in the Department of Engineering, Università degli Studi di Perugia, Italy
  May 2026

- Member of the final evaluation committee for the degree of Doctor of Philosophy, Doctoral programme in Industrial and Information Engineering, Università degli Studi di Perugia, Italy Apr. 2026
- External reviewer for PhD dissertation, Inter-university Doctoral Programme in Bioengineering and Medical-Surgical Sciences, Politecnico di Torino, Università di Torino, Italy Jan. 2026
- Member of the final evaluation committee for the degree of Doctor of Philosophy, Doctoral programme in Artificial Intelligence, Politecnico di Torino, Italy Jun. 2025
- Reviewer for Proyectos Sinérgicos I+D 2024, Comunidad de Madrid, Spain May 2025
- External reviewer for PhD dissertation, Doctoral Programme in Aerospace Science and Technology, Università degli Studi di Bologna, Italy Jan. 2024
- External reviewer for PhD dissertation, Doctoral Programme in Industrial Engineering, Università degli Studi di Firenze, Florence, Italy Dec. 2024
- External reviewer for PhD dissertation, Doctoral programme in Environment Design Innovation, University of Campania 'Luigi Vanvitelli', Italy, Sep. 2024 Sep. 2024
- External reviewer for PhD dissertation, EIDO International Doctoral School, Universidade de Vigo, Spain May 2024
- Member of the final evaluation committee for the degree of Doctor of Philosophy, Department of Industrial Engineering, Università degli Sudi di Firenze, Italy Apr. 2024
- Member of the final evaluation committee for the degree of Doctor of Philosophy, department of Industrial and Information Engineering, Università di Pavia, Italy Mar. 2024
- External reviewer for PhD dissertation, EIDO International Doctoral School, Universidade de Vigo, Spain Mar. 2024
- External reviewer for PhD dissertation, Doctoral Programme in Industrial Engineering, Università degli Studi di Firenze, Florence, Italy Dec. 2023
- External reviewer for PhD dissertation, International Doctoral School, Universidade de Vigo, Spain November 2023
- External reviewer for PhD dissertation, Department of Informatics, Systems and Communications, Università degli Studi di Milano-Bicocca Milan, Italy Nov. 2023
- Member of the final evaluation committee for the degree of Doctor of Philosophy, International Doctoral School, Universidade de Vigo, Spain Feb. 2023
- External reviewer for PhD dissertation, International Doctoral School, Universidade de Vigo, Spain April 2022
- Member of the final evaluation committee for the degree of Philosophiae Doctor, Faculty of Information Technology and Electrical Engineering, Norges Teknisk-Naturvitenskapelige Universitet (NTNU), Gjøvik, Norway Jan. 2022
- External reviewer for PhD dissertation, Department of Informatics, Systems and Communications, Università degli Studi di Milano-Bicocca Milan, Italy, Nov. 2021

- External reviewer for the Italian Ministry for University and Research, National Research Projects (PRIN) call 2020 May-Sep. 2021
- External reviewer for PhD dissertation, School of Electronic Engineering and Computer Science, Queen Mary University of London, United Kingdom May 2020
- External reviewer for PhD dissertation, Anna University, Chennai, India Feb. 2020
- External referee for academic promotions at London South Bank University, United Kingdom Feb. 2020
- External reviewer for PhD dissertation, University of Florence, Italy Dec. 2019
- Head of the evaluation panel for Fondimpresa Lifelong Learning projects ('Technology and/or digital innovation' call 4/2018), Rome, Italy Mar.–Sep. 2019
- External reviewer for the Young Reseachers Programme 'Rita Levi Montalcini', Italian Ministry of Education 2016
- Member of the final evaluation committee for the degree of Doctor of Philosophy, International Doctoral Programme, University of Vigo, Spain Jul. 2019
- Member of the final evaluation committee for the degree of Doctor of Philosophy, Department of Engineering, University of Almería, Spain Jul. 2013
- Member of the final evaluation committee for the degree of Doctor of Philosophy, Department of Mining and Environmental Engineering, University of Vigo, Spain Jul. 2011
- Member of the final evaluation committee for the degree of Doctor of Philosophy, Faculty of Sciences, University of Vigo, Spain Mar. 2010

Honours and awards

- Best paper award: G. Pascoletti, C. Di Natali, M. Sposito, G. Franceschini, F. Bianconi and E.M. Zanetti; A multibody model of the exoskeleton-human body system for ergonomic design, ADM2024 International Conference, Palermo, Italy Sep. 2024
- First Prize: S. Cascianelli, C. Tranfaglia, M.L. Fravolini, F. Bianconi, M. Minestrini, S. Nuvoli, N. Tambasco, M.E. Dottorini and B. Palumbo; *Right putamen and age are the most discriminant features to diagnose Parkinson's disease by using 123I-FP-CIT brain SPECT data by using an artificial neural network classifier, a classification tree (CIT)*, 4th International Medical Olympiad; Thessaloniki, Greece Nov. 2017
- Best paper award: R. Bello-Cerezo, F. Bianconi, S. Cascianelli, F. Di Maria, M. Fravolini and F. Smeraldi; *Hand-designed local image descriptors vs. off -the-shelf CNN-based features for texture classification: an experimental comparison*, 9th International Conference on Intelligent Interactive Multimedia: Systems and Services; Vilamoura, Portugal Jun. 2017
- Best paper award: M.J. Álvarez, A. Fernández, E. González, F. Bianconi, F.J. Aguilar and J. Armesto; *Image segmentation using multilayer Coordinated Cluster Representation*, International Joint Congress XXI INGEGRAF - XVII ADM; Lugo, Spain Jun. 2009

Patents and inventions

- IT patent no. 1413266: *Apparatus to acquire a plurality of superficial images of at least one body and related method*. Assignee: Mondial Marmi S.r.l.; inventors: F. Bianconi, S.A. Saetta, A. Fernández and E. González 2015.

Publications

Selected (sorted by date, newest first)

F. Bianconi, I. Palumbo, M.L. Fravolini, R. Chiari, M. Minestrini, L. Brunese, and B. Palumbo. Texture analysis on [18F]FDG PET/CT in non-small-cell lung cancer: Correlations between PET features, CT features, and histological types. *Molecular Imaging and Biology*, 21(6):1200–1209, December 2019

F. Bianconi, C. Cusano, P. Napoletano, and R. Schettini. CNN-based refactoring of hand-designed filters for texture analysis: A classic revisited. *IEEE Access*, 7:173076–173085, November 2019. Art. no. 8918055

F. Bianconi and E. González. Counting local n-ary patterns. *Pattern Recognition Letters*, 117:24–29, January 2019

J.N. Kather, C.-A. Weis, F. Bianconi, S.M. Melchers, L.R. Schad, T. Gaiser, A. marx, and F.G. Zöllner. Multi-class texture analysis in colorectal cancer histology. *Scientific Reports*, 6, June 2016. art. no. 27988

F. Bianconi and A. Fernández. On the occurrence probability of local binary patterns: A theoretical study. *Journal of Mathematical Imaging and Vision*, 40(3):259–268, July 2011

F. Bianconi, R. Harvey, P. Southam, and Fernández. Theoretical and experimental comparison of different approaches for color texture classification. *Journal of Electronic Imaging*, 20(4), October 2011. Article number 043006

Books

[1] F. Bianconi. *Data and process visualisation for graphic communication: A hands-on approach with Python*. Springer Nature, Cham, first edition, May 2024. XIX, 242 p., 14 b/w illustrations, 84 illustrations in colour

Edited books

[1] F. Bianconi, A. Fernández, and R.E. Sánchez-Yáñez, editors. *Texture and color in image analysis*. MDPI, 2021. Printed edition of the Special Issue Texture and Colour in Image Analysis, published in Applied Sciences

Book chapters

[1] F. Bianconi and A. Fernández. A unifying framework for LBP and related methods. In S. Brahnam, L. C. Jain, L. Nanni, and A. Lumini, editors, *Local binary patterns: New variants and applications*, volume 506 of *Studies in Computational Intelligence*, pages 17–46. Springer, 2014

[2] J.N. Kather, R. Bello-Cerezo, F. Di Maria, G.W. van Pelt, W.E. Mesker, N. Halama, and F. Bianconi. Classification of tissue regions in histopathological images: Comparison between pre-trained convolutional neural networks and local binary patterns variants. In L. Nanni, S. Brahnam, R. Brattin, S. Ghidoni, and L.C. Jain, editors, *Deep learners and deep learner descriptors for medical applications*, volume 186 of *Intelligent Systems Reference Library*, chapter 3, pages 95–115. Springer, 2020

Journal papers

[1] M. Scialpi, G. A. Antogiovanni, F. Bianconi, B. Palumbo, G. Bellezza, M. Mandarano, S. Ammar, G. B. Scalera, and R. Nicola. Quantitative assessment of prostate cancer using an individualized apparent diffusion coefficient intensity scale: a proposal. *Quantitative Imaging in Medicine and Surgery*, 16(8):664, jul 2026

[2] L. Filippi, F. Bianconi, A. Marongiu, A. Spanu, and B. Palumbo. SPECT-based radiomics in oncology: current evidence, technical challenges, and future directions. *Clinical and Translational Imaging*, 14:93–101, February 2026

[3] L. Filippi, F. Bianconi, V. Frantellizzi, C. Ferrari, A. Marongiu, M. S. De Feo, C. Battisti, G. Aghakhanyan, M. Gazzilli, N. Urbano, S. Nuvoli, D. Volterrani, M. L. Fravolini, G. Rubini, G. De Vincentis, A. Spanu, and B. Palumbo. Machine learning for automated differentiation of parkinson's disease and its mimics using ^{123}I-mIBG scintigraphy: insights from a multicentre real-world cohort (ITA-mIBG study). *European Journal of Nuclear Medicine and Molecular Imaging*, 53:3469–3480, January 2026

[4] B. Palumbo, L. Filippi, A. Marongiu, F. Bianconi, M. L. Fravolini, R. Danieli, V. Frantellizzi, G. De Vincentis, A. Spanu, and S. Nuvoli. Diagnostic accuracy of DaTQUANT® versus BasGanV2™ for ^{123}I-Ioflupane brain SPECT: A machine learning-based differentiation of Parkinson's disease and essential tremor. *Biomedicines*, 13(10), 2025. Art. no. 2367

[5] A. Marongiu, A. Spanu, B. Palumbo, F. Bianconi, L. Filippi, G. Madeddu, and S. Nuvoli. Artificial intelligence in PET imaging for Alzheimer's disease: A narrative review. *Brain Sciences*, 15(10), September 2025. Art no. 1038

[6] F. Bianconi, M. U. Khan, H. Du, and S. Jassim. Experimental assessment of conventional features, CNN-based features and ensemble schemes for discriminating benign versus malignant lesions on breast ultrasound images. *Ultrasonic Imaging*, 47(6):256–269, November 2025

[7] G. Pascoletti, C. Di Natali, D. G. Caldwell, M. Sposito, F. Bianconi, and E. M. Zanetti. The exoskeleton-human body system: a multibody model for fitting design analyses. *International Journal on Interactive Design and Manufacturing (IJIDeM)*, 2025. Published: 14 August 2025

[8] L. Filippi, F. Bianconi, C. Ferrari, F. Linguanti, C. Battisti, N. Urbano, M. Minestrini, S. G. Messina, L. Buci, A. Baldoncini, G. Rubini, O. Schillaci, and B. Palumbo. Digital versus analogue PET in parathyroid imaging: comparison of PET metrics and machine learning-based characterisation of hyperfunctioning lesions (the DIGI-PET study). *European Journal of Nuclear Medicine and Molecular Imaging*, 53:2122–2130, August 2025

[9] O. Bagni, R. Danieli, F. Bianconi, B. Palumbo, and L. Filippi. Clinical performance of analog and digital 18F-FDG PET/CT in pediatric epileptogenic zone localization: preliminary results. *Biomedicines*, 13(8), 2025. Art. no. 1887

[10] L. Filippi, F. Bianconi, M. Minestrini, G. De Vincentis, V. Frantellizzi, A. Marongiu, C. Ferrari, A. Giordano, D. D'Ambrosio, M. Gazzilli, F. Bertagna, O. Bagni, S. Panareo, P. Guglielmo, A. Lazzarato, M. Casali, G. Aghakhanyan, A. Spanu, L. Evangelista, O. Schillaci, and B. Palumbo. Multi-centre data harmonisation applied to heart-to-mediastinum quantification in parkinsonism (ITA-MIBG): a cross-calibration phantom study with tube and bottle. *Clinical and Translational Imaging*, 13:185–186, February 2025

[11] L. Filippi, F. Bianconi, O. Schillaci, and B. Palumbo. This union must happen: Radiomics and LAFOV PET/CT as the power couple of nuclear medicine. *European Journal of Nuclear Medicine and Molecular Imaging*, December 2024. Letter to the Editor

[12] F. Bianconi, R. Salis, M.L. Fravolini, M.U. Khan, L. Filippi, A. Marongiu, S. Nuvoli, A. Spanu, and B. Palumbo. Radiomics features from Positron Emission Tomography with [18F] Fluorodeoxyglucose can help predict cervical nodal status in patients with head and neck cancer. *Cancers*, 16(22), November 2024. Art. no. 3759

[13] T. B. Irshad, G. Pascoletti, F. Bianconi, and E. M. Zanetti. Mandibular bone segmentation from CT scans: Quantitative and qualitative comparison among software. *Dental Materials*, 40(8):e11–e22, 2024

[14] F Bianconi, R Salis, M. L. Fravolini, M. U. Khan, M. Minestrini, L. Filippi, A. Marongiu, S. Nuvoli, A. Spanu, and B. Palumbo. Performance analysis of six semi-automated tumour delineation methods on [18F] fluorodeoxyglucose positron emission tomography/computed tomography (FDG PET/CT) in patients with head and neck cancer. *Sensors*, 23(18), 2023. Art. no. 7952

[15] E. Buchicchio, A. De Angelis, F. Santoni, P. Carbone, F. Bianconi, and F. Smeraldi. Battery SOC estimation from EIS data based on machine learning and equivalent circuit model. *Energy*, 283, November 2023. Art. no. 128461

[16] L. Evangelista, F. Fiz, R. Laudicella, F. Bianconi, A. Castello, P. Guglielmo, V. Liberini, L. Manco, V. Frantellizzi, A. Giordano, L. Urso, S. Panareo, B. Palumbo, and L. Filippi. PET radiomics and response to immunotherapy in lung cancer: A systematic review of the literature. *Cancers*, 15(12), 2023. Art. no. 3258

[17] F. Bianconi, M. L. Fravolini, and B. Palumbo. Size measurement of lung nodules on CT: which diameter is most stable to inter-observer variability? *Clinical Imaging*, 99:38–40, July 2023

[18] L. Filippi, L. Urso, F. Bianconi, B. Palumbo, M. C. Marzola, L. Evangelista, and O. Schillaci. Radiomics and theranostics with molecular and metabolic probes in prostate cancer: toward a personalized approach. *Expert Review of Molecular Diagnostics*, 23(3):243–255, 2023

[19] E. Buchicchio, F. Bianconi, F. Smeraldi, A. De Angelis, F. Santoni, and P. Carbone. Battery state of charge estimation using machine learning and electrochemical impedance spectroscopy measurements. *Science Talks*, 4, December 2022. Art no. 100100

[20] E. Buchicchio, A. De Angelis, F. Santoni, P. Carbone, F. Bianconi, and F. Smeraldi. LiBEIS: A software tool for broadband electrochemical impedance spectroscopy of lithium-ion batteries. *Software Impacts*, 14, November 2022. Art. no. 100447

[21] S. Nuvoli, B. Palumbo, A. Marongiu, F. Bianconi, and A. Spanu. 123I-MIBG cardiac scintigraphy and heart/mediastinum ratio in neurodegenerative disorders: Is delayed scan really necessary? *Current Radiopharmaceuticals*, 15(4):257–258, December 2022

[22] S. Nuvoli, F. Bianconi, M. Rondini, A. Lazzarato, A. Marongiu, M.L. Fravolini, S. Cascianelli, S. Amici, L. Filippi, A. Spanu, and B. Palumbo. Differential diagnosis of Alzheimer disease vs mild cognitive impairment based on left temporal lateral lobe hypomethabolism on ^{18}F-FDG PET/CT and automated classifiers. *Diagnostics*, 12(10), September 2022. Art. no. 2425

[23] E. Buchicchio, A. De Angelis, F. Santoni, P. Carbone, F. Bianconi, and F. Smeraldi. Dataset on broadband electrochemical impedance spectroscopy of Lithium-Ion batteries for different values of the state-of-charge. *Data in Brief*, 45, 2022. Art. no 108589

[24] F. Bianconi, I. Palumbo, M. L. Fravolini, M. Rondini, M. Minestrini, G. Pascoletti, S. Nuvoli, A. Spanu, M. Scialpi, C. Aristei, and B. Palumbo. Form factors as potential imaging biomarkers to differentiate benign vs. malignant lung lesions on CT scans. *Sensors*, 22(13), 2022. Art no. 5044

[25] L. Filippi, F. Bianconi, O. Schillaci, A. Spanu, and B. Palumbo. The role and potential of 18F-FDG PET/CT in malignant melanoma: prognostication, monitoring response to targeted and immunotherapy, and radiomics. *Diagnostics*, 12(4), April 2022. Art. no. 929

[26] B Palumbo, F. Bianconi, M.L. Fravolini, I. Palumbo, S. Nuvoli, and A. Spanu. PET con trazadores de aminoácidos y radiómica: una oportunidad emergente para el diagnóstico de tumores cerebrales [PET with amino acid tracers and radiomics: An emerging opportunity for the diagnosis of brain tumours]. *Revista Española de Medicina Nuclear e Imagen Molecular*, 41(2):136, March 2022

[27] F. Bianconi and E. Brugnoli. Enumerating necklaces with transitions. *Bulletin of the Australian Mathematical Society*, 105(11):1–11, 2022

[28] F. Bianconi, A. Fernández, F. Smeraldi, and G. Pascoletti. Colour and texture descriptors for visual recognition: A historical overview. *Journal of Imaging*, 7(11), November 2021. Art. no. 245

[29] E. Chirikhina, A. Chirikhin, S. Dewsbury-Ennis, F. Bianconi, and P. Xiao. Skin characterizations by using contact capacitive imaging and high-resolution ultrasound imaging with machine learning algorithms. *Applied Sciences*, 11(18), September 2021. Art. no. 8714

[30] B. Palumbo, F. Bianconi, and I. Palumbo. Solitary pulmonary nodule: Is positron emission tomography/computed tomography radiomics a valid diagnostic approach? *Lung India*, 38(5):405–407, September 2021. Editorial

[31] F. Bianconi, M.L. Fravolini, I. Palumbo, G. Pascoletti, S. Nuvoli, M. Rondini, A. Spanu, and B. Palumbo. Impact of lesion delineation and intensity quantisation on the stability of texture features from lung nodules on CT: A reproducible study. *Diagnostics*, 11(7), July 2021. Art. no. 1224

[32] F. Bianconi, M.L. Fravolini, S. Pizzoli, I. Palumbo, M. Minestrini, M. Rondini, S. Nuvoli, A. Spanu, and B. Palumbo. Comparative evaluation of conventional and deep learning methods for semi-automated segmentation of pulmonary nodules on CT. *Quantitative Imaging in Medicine and Surgery*, 11(7):3286 – 3305, July 2021

[33] F. Bianconi, A. Fernández, and R.E. Sánchez-Yáñez. Special issue texture and color in image analysis. *Applied Sciences*, 11(9), April 2021. Art. no. 3801. Editorial

[34] B. Palumbo, F. Bianconi, S. Nuvoli, A. Spanu, and M.L. Fravolini. Artificial intelligence techniques support nuclear medicine modalities to improve the diagnosis of Parkinson's disease and Parkinsonian syndromes. *Clinical and Translational Imaging*, 9(1):19–35, February 2021

[35] F. Bianconi, J.N. Kather, and C.C. Reyes-Aldasoro. Experimental assessment of color deconvolution and color normalization for automated classification of histology images stained with hematoxylin and eosin. *Cancers*, 12(11), November 2020. Art. no. 3337

[36] B. Palumbo, F. Bianconi, I. Palumbo, M.L. Fravolini, M. Minestrini, S. Nuvoli, M.L. Stazza, M. Rondini, and A. Spanu. Value of shape and texture features from 18F-FDG PET/CT to discriminate between benign and malignant solitary pulmonary nodules: An experimental evaluation. *Diagnostics*, 10, September 2020. Art no. 696

[37] E. Chirikhina, A. Chirikhin, P. Xiao, S. Dewsbury-Ennis, and F. Bianconi. In vivo assessment of water content, trans-epidermialwater loss and thickness in human facial skin. *Applied Sciences*, 10(17), September 2020. Art. no. 6139

[38] C. Buratti, E. Belloni, F. Merli, and F. Bianconi. Experimental characterization of the color rendering properties of transparent monolithic aerogel. *Solar Energy*, 205:183–191, July 2020

[39] B. Palumbo, R. Capozzi, F. Bianconi, M.L. Fravolini, S. Cascianelli, S.G. Messina, G. Bellezza, A. Sidoni, F. Puma, and M. Ragusa. Classification model to estimate MIB-1 (Ki 67) proliferation index in NSCLC patients evaluated with 18F-FDG-PET/CT. *Anticancer Research*, 40(6):3355–3360, June 2020

[40] S. Nuvoli, A. Spanu, M.L. Fravolini, F. Bianconi, S. Cascianelli, G. Madeddu, and B. Palumbo. [123I]metaiodobenzylguanidine (MIBG) cardiac scintigraphy and automated classification techniques in Parkinsonian disorders. *Molecular Imaging and Biology*, 22(3):703–710, June 2020

[41] F. Bianconi, I. Palumbo, A. Spanu, S. Nuvoli, M.L. Fravolini, and B. Palumbo. PET/CT radiomics in lung cancer: An overview. *Applied Sciences*, 5(10), March 2020. Art. no. 1718

[42] F. Smeraldi, F. Bianconi, A. Fernández, and E. González. Partial order rank features in colour space. *Applied Sciences*, 10(2), jan 2020. Art. no. 499

[43] F. Bianconi, I. Palumbo, M.L. Fravolini, R. Chiari, M. Minestrini, L. Brunese, and B. Palumbo. Texture analysis on [18F]FDG PET/CT in non-small-cell lung cancer: Correlations between PET features, CT features, and histological types. *Molecular Imaging and Biology*, 21(6):1200–1209, December 2019

[44] F. Bianconi, C. Cusano, P. Napoletano, and R. Schettini. CNN-based refactoring of hand-designed filters for texture analysis: A classic revisited. *IEEE Access*, 7:173076–173085, November 2019. Art. no. 8918055

[45] M. Scialpi, F. Bianconi, V. Cantisani, and B. Palumbo. Radiomic machine learning: Is it really a useful method for the characterization of prostate cancer? *Radiology*, 291(1):269–270, April 2019

[46] F. Bianconi and E. González. Counting local n-ary patterns. *Pattern Recognition Letters*, 117:24–29, January 2019

[47] F. Bianconi, M.L. Fravolini, R. Bello-Cerezo, M. Minestrini, M. Scialpi, and B. Palumbo. Evaluation of shape and textural features from CT as prognostic biomarkers in non-small cell lung cancer. *Anticancer Research*, 38(4):2155–2160, April 2018

[48] F. Bianconi, R. Bello-Cerezo, and P. Napoletano. Improved opponent color local binary patterns: an effective local image descriptor for color texture classification. *Journal of Electronic Imaging*, 27(1), December 2018. Art. No. 011002

[49] S. Cascianelli, C. Tranfaglia, M.L. Fravolini, F. Bianconi, M. Minestrini, S. Nuvoli, N. Tambasco, M.E. Dottorini, and B. Palumbo. Right putamen and age are the most discriminant features to diagnose parkinson's disease by using ^{123}I-FP-CIT brain SPET data by using an artificial neural network classifier, a classification tree (CIT). *Hellenic journal of nuclear medicine*, 20:165, 2017

[50] F. Bianconi, E. Chirikhina, F. Smeraldi, C. Bontozoglou, and P. Xiao. Personal identification based on skin texture features from the forearm and multi-modal imaging. *Skin Research and Technology*, 23(3):392–398, August 2017

[51] F. Di Maria, M. Barratta, F. Bianconi, P. Placidi, and D. Passeri. Solid anaerobic digestion batch with liquid digestate recirculation and wet anaerobic digestion of organic waste: Comparison of system performances and identification of microbial guilds. *Waste Management*, 59:172–180, January 2017

[52] R. Bello-Cerezo, F. Bianconi, A. Fernández, E. González, and F. Di maria. Experimental comparison of color spaces for material classification. *Experimental comparison of color spaces for material classification*, 25(6), November 2016. Art. no. 061406

[53] E. González, F. Bianconi, and A. Fernández. An investigation on the use of local multi-resolution patterns for image classification. *Information Sciences*, 361-362:1–13, September 2016

[54] J.N. Kather, C.-A. Weis, F. Bianconi, S.M. Melchers, L.R. Schad, T. Gaiser, A. marx, and F.G. Zöllner. Multi-class texture analysis in colorectal cancer histology. *Scientific Reports*, 6, June 2016. art. no. 27988

[55] M.A. Aguilar, A. Fernández, F.J. Aguilar, F. Bianconi, and A. García Lorca. Classification of urban areas from geoeye-1 imagery through texture features based on histograms of equivalent patterns. *European Journal of Remote Sensing*, 49:93–120, March 2016

[56] F. Di Maria, F. Bianconi, C. Micale, S. Baglioni, and M. Marionni. Quality assessment for recycling aggregates from construction and demolition waste: An image-based approach for particle size estimation. *Waste Management*, 344-352:344–352, February 2016

[57] F. Bianconi, F. Di maria, C. Micale, A. Fernández, and R.W. Harvey. Grain-size assessment of fine and coarse aggregates through bipolar area morphology. *Machine Vision and Applications*, 26(6):775–789, August 2015

[58] F. Bianconi, E. González, and A. Fernández. Dominant local binary patterns for texture classification: Labelled or unlabelled? *Pattern Recognition Letters*, 65:8–14, July 2015

[59] L. Postrioti, G. Brizi, C. Ungaro, M. Mosser, and F. Bianconi. A methodology to investigate the behaviour of urea-water sprays in high temperature air flow for SCR de-NOx applications. *Fuel*, 150:548–557, June 2015

[60] F. Bianconi, A. Fernández, and A. Álvarez Larrán. Discrimination between tumour epithelium and stroma via perception-based features. *Neurocomputing*, 154:119–126, April 2015

[61] E. González, A. Fernández, and F. Bianconi. General framework for rotation invariant texture classification through co-occurrence of patterns. *Journal of Mathematical Imaging and Vision*, 50(3):300–313, November 2014

[62] F. Bianconi and A. Fernández. Rotation invariant co-occurrence features based on digital circles and discrete Fourier transform. *Pattern Recognition Letters*, 48:34–41, October 2014

[63] F. Bianconi and A. Fernández. An appendix to "texture databases – A comprehensive survey". *Pattern Recognition Letters*, 45(1):33–38, August 2014

[64] F. Bianconi, L. Ceccarelli, A. Fernández, and S.A. Saetta. A sequential machine vision procedure for assessing paper impurities. *Computers in Industry*, 65(2):325–332, February 2014

[65] M.A. Aguilar, F. Bianconi, F.J. Aguilar, and I. Fernández. Object-based greenhouse classification from GeoEye-1 and WorldView-2 stereo imagery. *Remote Sensing*, 6(5):3554–3582, May 2014

[66] V. Caldarelli, L. Ceccarelli, F. Bianconi, S.A. Saetta, and A. Fernández. Performance assessment of multi-level image thresholding for paper quality inspection. *International Journal of Service and Computing Oriented Manufacturing*, 1(4):281–294, 2014

[67] E. González, F. Bianconi, M.X. Álvarez, and S.A. Saetta. Automatic characterization of the visual appearance of industrial materials through colour and texture analysis: An overview of methods and applications. *Advances in Optical Technologies*, 2013. Art. no. 503541

[68] F. Bianconi, A. Fernández, E. González, and S.A. Saetta. Performance analysis of colour descriptors for parquet sorting. *Expert Systems with Applications*, 40(5):1636–1644, April 2013

[69] A. Fernández, M. X. Álvarez, and F. Bianconi. Texture description through histograms of equivalent patterns. *Journal of Mathematical Imaging and Vision*, 45(1):76–102, January 2013

[70] F. Bianconi, E. González, A. Fernández, and S.A. Saetta. Automatic classification of granite tiles through colour and texture features. *Expert Systems with Applications*, 39(12):11212–11218, September 2012

[71] A. Fernández, O. Ghita, E. González, F. Bianconi, and P.F. Whelan. Evaluation of robustness against rotation of LBP, CCR and ILBP features in granite texture classification. *Machine Vision and Applications*, 22(6):913–926, November 2011

[72] F. Bianconi, R. Harvey, P. Southam, and Fernández. Theoretical and experimental comparison of different approaches for color texture classification. *Journal of Electronic Imaging*, 20(4), October 2011. Article number 043006

[73] F. Bianconi and A. Fernández. On the occurrence probability of local binary patterns: A theoretical study. *Journal of Mathematical Imaging and Vision*, 40(3):259–268, July 2011

[74] A. Fernández, Álvarez M.X., and F. Bianconi. Image classification with binary gradient contours. *Optics and Lasers in Engineering*, 49(9-10):1177–1184, September 2011

[75] M.J. Álvarez, E. González, F. Bianconi, J. Armesto, and A. Fernández. Colour and texture features for image retrieval in granite industry. *DYNA (Colombia)*, 77(161):121–130, January 2010

[76] M.J. Álvarez, A. Fernaández, E. González, F. Bianconi, F.J. Aguilar, and J. Armesto. Image segmentation using rotation-invariant multilayer coordinated clusters representation. *Anales de Ingeniería Gráfica*, 21:7–14, 2010

[77] E. González, F. Bianconi, and A. Fernaández. A comparative review of colour features for content-based image retrieval. *Anales de Ingeniería Gráfica*, 21:7–14, 2010

[78] F. Bianconi, A. Fernández, E. González, and J. Armesto. Robust color texture features based on ranklets and discrete Fourier transform. *Journal of Electronic Imaging*, 18(4), 2009. Art. no. 043012

[79] F. Bianconi, A. Fernández, E. González, D. Caride, and A. Calviño. Rotation-invariant colour texture classification through multilayer CCR. *Pattern Recognition Letters*, 30(8):765–773, June 2009

[80] F. Bianconi and A. Fernández. Evaluation of the effects of gabor filter parameters on texture classification. *Pattern Recognition*, 40(12):3325–3335, December 2007

[81] F. Bianconi, S.A. Saetta, and L. Tiacci. A web-based simulation game as a learning tool for the design process of complex systems. *Journal of Design Research*, 5(2):253–272, 2006

[82] P. Di Stefano, F. Bianconi, and L. Di Angelo. An approach for feature semantics recognition in geometric models. *CAD Computer Aided Design*, 36(10):993–1009, September 2004

[83] F. Bianconi. Bridging the gap between CAD and CAE using STL files. *International Journal of CAD/CAM*, 2(1):55–67, 2002

Conference proceedings

[1] A. Fernández and F. Bianconi. Designing effective bar charts for global health data. In I. Alvarez Gonzalez, X. Amezua Lasuen, M. Iturrondobeitia Ellacuria, R. Minguez Gabiña, E. Rodriguez Hortelano, I. San Sebastian Ibirriaga, and J. S. Velázquez Blázquez, editors, *Proceedings of the 35th INGEGRAF International Technical Conference 'Design for one health'*, pages 351–375, Bilbao, Spain, September 2026. EHU (Servicio de Publicaciones de la Universidad del País Vasco)

[2] F. Bianconi, M. L. Fravolini, C. Buratti, G. Pascoletti, and E. M. Zanetti. Identification of the size distribution of SEM particles by conventional texture descriptors and deep features from pre-trained convolutional networks. In G. Berselli, A. O. Andrisano, P. Di Stefano, C. Rizzi, and F. Gherardini, editors, *Proceedings of the Fifth International Conference on Design Tools and Methods in Industrial Engineering, ADM 2025, September 3–5, 2025, Genova, Italy*, volume Advances on Design Engineering V–Volume II: Innovations in Products, Advanced Manufacturing and Design Engineering of *Lecture Notes in Mechanical Engineering*, pages 169–180. Springer, 2026

[3] G. Pascoletti, T. B. Irshad, G. Putame, F. Bianconi, and E. M. Zanetti. Temporomandibular joint prosthesis design for enhanced stability and mobility. In G. Berselli, A. O. Andrisano, P. Di Stefano, C. Rizzi, and F. Gherardini, editors, *Proceedings of the Fifth International Conference on Design Tools and Methods in Industrial Engineering, ADM 2025, September 3–5, 2025, Genova, Italy*, volume Advances on Design Engineering V–Volume II: Innovations in Products, Advanced Manufacturing and Design Engineering of *Lecture Notes in Mechanical Engineering*, pages 3–15. Springer, 2026

[4] A. Fernández, L. Díaz-Vilariño, and F. Bianconi. AutoCAD automation through Python scripting. In C. Manchado del Val, R. Mirálbes Buil, C. Torrecillas, and M. Morato-Moreno, editors, *Proceedings of the 34th INGEGRAF International Conference, INGEGRAF2025, Seville, Spain, 25–27 June 2025*, volume Design Tools and Methods in Industrial Engineering V Design Tools and Methods in Industrial Engineering V of *Lecture Notes in Mechanical Engineering*, pages 169–179. Springer, 2026

[5] M. U. Khan, F. Bianconi, H. Du, and S. Jasssim. Hand-crafted vs. deep CNN features to distinguish benign from malignant lesions in breast ultrasound images. In *Proceedings of the 9th International Conference on Control, Automation and Diagnosis (ICCAD 2025)*, Barcelona, Spain, July 2025. IEEE

[6] G. Pascoletti, C. Di Natali, M. Sposito, G. Franceschini, F. Bianconi, and E. M. Zanetti. A multibody model of the exoskeleton-human body system for ergonomic design. In P. Di Stefano, F. Gherardini, V. Nigrelli, C. Rizzi, G. Sequenzia, and D. Tumino, editors, *Proceedings of the International Conference of the Italian Association of Design Methods and Tools for Industrial Engineering (ADM 2024)*, volume Design Tools and Methods in Industrial Engineering IV of *Lecture Notes in Mechanical Engineering*, pages 182–189, Palermo, Italy, September 2025. Springer

[7] F. Bianconi, M. L. Fravolini, E. Caltana, M. U. Khan, and B. Palumbo. Classification of lung nodules on CT via pseudo-colour images and deep features from pre-trained convolutional networks. In R. Schettini, A. Trémeau, S. Tominaga, S. Bianco, and M. Buzzelli, editors, *Computational Color Imaging. CCIW 2024.*, volume 15193 of *Lecture Notes in Computer Science*, page 221–232, Milan, Italy, September 2024. Springer, Cham

[8] M. U. Khan, F. Bianconi, M. L. Fravolini, and B. Palumbo. Sensitivity of radiomics features to region volume: A CT phantom study. In *Proceedings of the 8th International Conference on Control, Automation and Diagnosis (ICCAD 2024)*, Paris, France, May 2025. IEEE

[9] F. Bianconi, C. Buratti, and G. Pascoletti. Relationship between colour themes and perceived aesthetic quality in colour images: an exploratory study. In M. Carfagni, R. Furferi, P. Di Stefano, L. Governi, and F. Gherardini, editors, *Proceedings of the International Conference of the Italian Association of Design Methods and Tools for Industrial Engineering (ADM 2023)*, volume Design Tools and Methods in Industrial Engineering III of *Lecture Notes in Mechanical Engineering*, pages 409–417, Florence, Italy, September 2024. Springer

[10] M. Fì, E. M. Zanetti, F. Bianconi, and G. Pascoletti. From real-time acquisition to mesh morphing of foot at different positions. In M. Carfagni, R. Furferi, P. Di Stefano, L. Governi, and F. Gherardini, editors, *Proceedings of the International Conference of the Italian Association of Design Methods and Tools for Industrial Engineering (ADM 2023)*, volume Design Tools and Methods in Industrial Engineering III of *Lecture Notes in Mechanical Engineering*, pages 3–11, Florence, Italy, September 2024. Springer

[11] F. Bianconi, C. Cusano, P. Napoletano, and R. Schettini. Optimizing Gabor texture features for materials recognition by convolutional neural networks. In *London Imaging Meeting*, volume 4, London, United Kingdom, June 2023. Society for Imaging Science and Technology. Article ID: 26

[12] F. Bianconi, M.L. Fravolini, G. Pascoletti, I. Palumbo, M. Scialpi, C. Aristei, and B. Palumbo. Correlation between IBSI morphological features and manually-annotated shape attributes on lung lesions at CT. In G. Yang, A. Aviles-Rivero, M. Roberts, and C.-B. Schönlieb, editors, *Proceedings of the Annual Conference on Medical Image Understanding and Analysis, MIUA 2022*, volume 13413 of *Lecture Notes in Computer Science*, pages 767–777, Cambridge, United Kingdom, July 2022. Springer

[13] F. Bianconi, P. Conti, E. M. Zanetti, and G. Pascoletti. A benchmark of traditional visual descriptors and convolutional networks 'off-the-shelf' for anomaly detection. In S. Gerbino, A. Lanzotti, M. Martorelli, R. M. Buil, C. Rizzi, and L. Roucoules, editors, *Proceedings of the International Joint Conference on Mechanics, Design Engineering & Advanced Manufacturing, JCM 2022*, volume Advances on Mechanics, Design Engineering and Manufacturing IV of *Lecture Notes in Mechanical Engineering*, pages 533–545, Ischia, Italy, June 2022. Springer

[14] G. Pascoletti, F. Bianconi, P. Conti, and E. M. Zanetti. Design of a snap-fit joint through a multibody model. In S. Gerbino, A. Lanzotti, M. Martorelli, R. M. Buil, C. Rizzi, and L. Roucoules,

editors, *Proceedings of the International Joint Conference on Mechanics, Design Engineering & Advanced Manufacturing, JCM 2022*, volume Advances on Mechanics, Design Engineering and Manufacturing IV of *Lecture Notes in Mechanical Engineering*, pages 793–802, Ischia, Italy, June 2022. Springer

[15] F. Bianconi. Experimental analysis of colour constancy and colour augmentation for painting classification by artistic genre: preliminary results. In *Proceedings of the 2nd International Conference Florence Heri-Tech: The Future of Heritage Science and Technologies, HERITECH 2020*, volume 949 of *IOP Conference Series: Materials Science and Engineering*, Florence, Italy, October 2020. Art. no. 012065

[16] F. Bianconi, M.L. Fravolini, I. Palumbo, and B. Palumbo. Shape and texture analysis of radiomic data for computer-assisted diagnosis and prognostication: an overview. In C. Rizzi, A.O. Andrisano, F. Leali, F. Gherardini, F. Pini, and A. Vergnano, editors, *Proceedings of the International Conference on Design Tools and Methods in Industrial Engineering (ADM)*, Lecture Notes in Mechanical Engineering, pages 3–14, Modena, Italy, September 2019. Springer

[17] R. Bello-Cerezo, F. Bianconi, F. Di Maria, P. Napoletano, and F. Smeraldi. Comparative evaluation of hand-crafted image descriptors vs. off-the-shelf CNN-based features for colour texture classification under ideal and realistic conditions. *Applied Sciences*, 9(4), February 2019. Article number: 738

[18] F. Bianconi and F. Evaluation of colour pre-processing on patch-based classification of H&E-stained images. In C.C. Reyes-Aldasoro, A. Janowczyk, M. Veta, and P. Bankhead, editors, *Proceedings of the 15th European Congress on Digital Pathology, ECDP 2019*, volume 11435 of *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, pages 56–64, Warwick; United Kingdom, April 2019. Springer

[19] R. Bello-Cerezo, P. Fieguth, and F. Bianconi. LBP-motivated colour texture classification. In L. Leal-Taixé and S. Roth, editors, *Proceedings of the 2nd International Workshop on Compact and Efficient Feature Representation and Learning in Computer Vision (in conjunction with ECCV 2018)*, volume 11132 of *Lecture Notes in Computer Science*, pages 517–533, Münich, Germany, September 2018

[20] M. Abdollahyan, S. Cascianelli, E. Bellocchio, G. Costante, T. A. Ciarfuglia, F. Bianconi, F. Smeraldi, and M. Fravolini. Visual localization in the presence of appearance changes using the partial order kernel. In *Proceedings of the 26th European Signal Processing Conference (EUSIPCO)*, pages 702–706, Rome, Italy, September 2018

[21] F. Bianconi and R. Bello-Cerezo. Evaluation of visual descriptors for painting categorisation. In *Florence Heri-Tech – The Future of Heritage Science and Technologies*, volume 364 of *IOP Conference Series: Materials Science and Engineering*, Florence, Italy, May 2018. IOPScience. Art. no. 012037

[22] R. Bello-Cerezo, F. Bianconi, S. Cascianelli, M.L. Fravolini, F. Di maria, and F. Smeraldi. Hand-designed local image descriptors vs. off-the-shelf cnn-based features for texture classification: An experimental comparison. In *Proceedings of the 10th KES International Conference on Intelligent Interactive Multimedia Systems and Services, IIMSS 2017*, volume 76 of *Smart Innovation, Systems and Technologies*, pages 1–10. Springer, June 2018

[23] S. Cascianelli, R. Bello-Cerezo, F. Bianconi, M.L. Fravolini, M. Belal, B. Palumbo, and J.N. Kather. Dimensionality reduction strategies for cnn-based classification of histopathological images. In *Proceedings of the 10th KES International Conference on Intelligent Interactive Multimedia Systems and Services, IIMSS 2017*, volume 76 of *Smart Innovation, Systems and Technologies*, pages 21–30. Springer, June 2018

[24] A. Fernández, D. Lima, F. Bianconi, and F. Smeraldi. Compact color texture descriptor based on rank transform and product ordering in the RGB color space. In *Proceedings of the IEEE International Conference on Computer Vision Workshops, ICCVW 2017*, pages 1032–1040, Venice, Italy, October 2017. Institute of Electrical and Electronics Engineers

[25] F. Bianconi, F. Smeraldi, M. Abdollahyan, and P. Xiao. On the use of skin texture features for gender recognition: An experimental evaluation. In *Proceedings of the 6th International Conference on Image Processing Theory, Tools and Applications, IPTA*, Oulu, Finland, December 2017. Art. no. 7821018

[26] F. Bianconi, R. Bello-Cerezo, P. Napoletano, and F. Di maria. Improved opponent colour local binary patterns for colour texture classification. In S. Bianco, R. Schettini, S. Tominaga, and A. Tremeau, editors, *Proceedings of the 6th Computational Color Imaging Workshop (CCIW'17)*, volume 10213 of *Lecture Notes in Computer Science*, pages 272–281, Milan, Italy, March 2017. Springer

[27] J. Pardo-Balado, A. Fernández, and F. Bianconi. Texture classification using rotation invariant LBP based on digital polygons. In V. Murino and E. Puppo, editors, *New Trends in Image Analysis and Processing – ICIAP 2015 Workshops*, volume 9281 of *Lecture Notes in Computer Science*, pages 87–94, Genoa, Italy, September 2015. Springer

[28] F. Bianconi, R. Bello-Cerezo, A. Fernández, and E. González. On comparing colour spaces from a performance perspective: Application to automated classification of polished natural stones. In V. Murino and E. Puppo, editors, *New Trends in Image Analysis and Processing – ICIAP 2015 Workshops*, volume 9281 of *Lecture Notes in Computer Science*, pages 71–78, Genoa, Italy, September 2015. Springer

[29] L. Ceccarelli, F. Bianconi, S.A. Saetta, A. Fernández, and V. Caldarelli. Experimental comparison of image thresholding methods for defect detection in the papermaking process. In *Proceedings of the 25th European Modeling & Simulation Symposium (EMSS)*, Athens, Greece, September 2013

[30] R. Harrison, F. Bianconi, R. Harvey, and W. Wang. A texture analysis approach to identifying sabellaria spinulosa colonies in sidescan sonar imagery. In *Proceedings of the 15th Irish Machine Vision and Image Processing Conference (IMVIP)*, pages 58–63, Dublin, Ireland, September 2011. Conference Publishing Services. Art. no. 6167881

[31] F. Bianconi, P. Conti, and G. Pazzaglia. Semi-automatic modeling of reverse-engineered shapes through design-by-feature and genetic algorithms. In G. Psihoyios and T.E Simos, editors, *Proceedings of the International e-Conference on Computer Science 2007, IeCCS*, volume 1060 of *AIP Conference Proceedings*, pages 125–128. American Institute of Physics Inc., November 2008

[32] S. Moroni, F. Bianconi, and P. Conti. Concurrent evaluation of manufacturing cost in product design and simulation. In *Proceedings of the International Mediterranean Modeling Multiconference, I3M*, Genoa, Italy, October 2007

[33] F. Bianconi, P. Conti, and S. Moroni. An approach to multidisciplinary product modeling and simulation through design-by-feature and classification trees. In F. De Felice, editor, *Proceedings of the 16th IASTED International Conference on Applied Simulation and Modelling, ASM*, pages 288–293, Palma de Mallorca, Spain, August 2007. ACTA Press

[34] F. Bianconi, A. Fernández, E. González, and F. Ribas. Texture classification through combination of sequential colour texture classifiers. In D. Rueda, D. Mery, and J. Kittler, editors, *Progress in Pattern Recognition, Image Analysis and Applications. 12th Iberoamerican Congress on Pattern Recognition, CIARP 2007.*, volume 4756 of *Lecture Notes in Computer Science*, pages 231–240, Valparaíso, Chile, nov 2007. Springer

[35] F. Bianconi and P. Conti. On the evaluation of sampling density distribution of a camera. In J.J. Villanueva, editor, *Proceedings of the 6th IASTED International Conference on Visualization, Imaging, and Image Processing, VIIP*, pages 19–24, Palma De Mallorca, Spain, August 2006. ACTA Press

[36] F. Bianconi, P. Conti, and L. Di Angelo. Interoperability among CAD/CAM/CAE systems: A review of current research trends. In *Proceedings of Geometric Modeling and Imaging New Trends*, pages 82–89, Lundon, United Kingdom, July 2006. IEEE Computer Society

[37] F. Bianconi, S.A. Saetta, L. Tiacci, and S. Viazzo. A web-based system to simulate a competitive design environment. In *Summer Computer Simulation Conferencs, SCSC – part of the Summer Simulation Multiconference, SummerSim*, pages 35–40, Philadelphia, PA, United States, July 2005. The Society for Modeling and Simulation International

[38] F. Bianconi, S.A. Saetta, and L. Tiacci. Simulation of a competitive product design environment for engineering education. In N. Callaos, editor, *Proceedings of the 9th World Multi-Conference on Systemics, Cybernetics and Informatics, WMSCI*, volume 10, pages 306–311, Orlando, FL, United States, July 2005. International Institute of Informatics and Systemics

[39] M. Argento, S. Barone, F. Bianconi, P. Conti, and E. Rosati. Reverse engineering and CFD analysis: A case study. In M.H. Hamza, editor, *Proceedings of the IASTED International Conference on Applied Simulation and Modelling*, pages 323–328, Rhodes, Greece, June 2004. ACTA Press. Art. no. 443-130

[40] F. Bianconi and P. Di Stefano. An intermediate level representation scheme for secondary features recognition and B-rep model simplification. In *Proceedings of Shape Modeling International, SMI*, pages 99–108, Seoul, Korea, May 2003. IEEE Computer Society. Art. no. 1199606

[41] F. Bianconi and P. Conti. Collaborative product modelling in heterogeneous environments: An approach based on XML Schema. In R. Jardim-Gonçalves, J. Cha, and A. Steiger-Garção, editors, *Proceedings of the 10th ISPE International Conference on Concurrent Engineering Research and Application*, volume 1: Enhanced Interoperable Systems, pages 303–310, Funchal, Portugal, July 2003. CRC Press

[42] F. Bianconi. A distributed object framework for collaborative CAD. In R. Jardim-Gonçalves, R. Roy, and A. Steiger-Garção, editors, *Proceedings of the 9th ISPE International Conference on Concurrent Engineering Research and Application*, pages 135–143, Cranfield, United Kingdom, July 2002. A.A. Balkema Publishers

Memberships & professional affiliations

- Member of the Italian Society of Nuclear Medicine and Molecular Imaging 2023–present
- Member of the Applied Geotechnologies Research Group ('TF1') at the Universdade de Vigo, Spain 2023–present
- Member of the research group *Gestión integrada del territorio y tecnologías de la información espacial* at the Universidad de Almería, Spain 2015–present
- Senior Member of the Institute of Electrical and Electronics Engineers (IEEE) 2013–present
- Member of the Institute of Electrical and Electronics Engineers (IEEE) 2000–2013
- Court-appointed Expert, Perugia Court of Justice, Perugia, Italy 1998–present
- Chartered Engineer, Perugia Engineering Council, Perugia, Italy 1997–present